RegimeLift: Targeted Residual Uplift Across Nested Gemma 3n Operating Regimes

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Abstract

This paper studies state uplift across nested operating regimes in Gemma 3n. Gemma 3n is distributed as separate E2B and E4B artifacts, but its MatFormer-style design makes the reduced E2B-like regime part of the same parent family rather than an unrelated checkpoint. That architectural fact motivates a narrower and more testable question than heterogeneous model bridging: can internal state from the reduced operating regime be corrected just enough for decoding to resume under the fuller regime with improved behavioral fidelity? We investigate this question using paired low/full runs on identical prompt prefixes, residual-state translators with null controls, live handoff metrics at horizons 1/4/8/16, and a sequence of kill-tests that isolate where intervention actually matters. The main empirical result is that broad residual training is insufficient, while a targeted residual uplift at `layer34,last1` produces strong continuation recovery on a heldout80 split. The resulting picture is clear: the dominant bottleneck is intervention placement and regime-specific control, not generic state-fitting capacity.

1 Introduction

The project in this repository began with a broader bridge hypothesis: can one transplant reasoning state across distinct language models? That hypothesis was too unconstrained. RegimeLift narrows the problem to a setting where structural relatedness is real and measurable: regime handoff inside Gemma 3n's shared E2B/E4B family.

Gemma 3n is documented as a nested / selective-activation architecture in which reduced and fuller operating modes coexist inside one parent family [2, 3]. The E2B and E4B artifacts are published separately, but the scientific object of interest here is not transfer between unrelated checkpoints. It is uplift between two operating regimes that were trained as part of the same nested family. That difference matters. If the reduced regime is an embedded sibling of the fuller regime, then a useful handoff might require only a small corrective transport map rather than a universal latent bridge.

This paper asks a deliberately limited question:

> Can a low-activation Gemma 3n prefix state be uplifted into the corresponding fuller-regime state so that decoding resumes closer to native full-regime behavior?

The answer is now evidence-based rather than speculative. Across completed separation tests, null controls, layer sweeps, suffix-span sweeps, and targeted-site studies, the work supports four concrete claims:

1. low/full regimes are genuinely separated;

2. identity is an exact control for the live handoff path;
3. local state alignment alone is not enough;
4. a specific learned intervention site, `layer34,last1`, yields the first robust continuation recovery result.

Contributions. The current draft makes four contributions.

- It formulates same-parent regime uplift in Gemma 3n as a falsifiable handoff problem rather than a generic manifold-alignment claim.
- It introduces a paired-state extraction and evaluation pipeline that measures both local alignment and resumed decoding behavior.
- It shows empirically that intervention site dominates broad translator capacity: broad MLP training improves local metrics, but targeted training at the correct site recovers behavior.
- It freezes a first practical learned handoff result at `layer34,last1` while documenting the active larger held-out validation as the next gate.

2 Background and framing

2.1 From archived He-LMAS to RegimeLift

The archived He-LMAS project attempted heterogeneous cross-model transfer. That historical work remains useful as context, but it is not the active research track [1]. RegimeLift changes the problem class:

- Archived He-LMAS: transfer between different models.
- RegimeLift: uplift between nested operating regimes of the same parent family.

That reframing is not cosmetic. It replaces an ambitious but weakly constrained hypothesis with a narrower one whose failure modes can be isolated experimentally.

2.2 Why Gemma 3n is the right testbed

Gemma 3n is explicitly documented as a nested family with selective activation and Mix-n-Match style operating regimes [2]. The reduced E2B-like path and the fuller E4B path are therefore related in a stronger sense than two arbitrary checkpoints. They are separate published artifacts, but they still belong to one shared structural family. For this paper, the important implication is:

> the reduced path is different enough to create a real handoff problem, yet related enough that a small corrective map is plausible.

That is the central scientific bet behind RegimeLift.

3 Related work and positioning

RegimeLift sits between two literatures. The first is standard transformer language modeling [4], which gives the baseline architecture and the residual-stream object we intervene on. The second is nested-representation or Matryoshka-style modeling [3], which makes same-family operating-mode comparisons scientifically plausible in the first place.

What RegimeLift does *not* claim is equally important. This is not a universal latent-alignment paper, not a cross-family checkpoint-bridging claim, and not evidence that one model's reasoning can always be transplanted into another. The archived He-LMAS line explored that broader space [1]. The current work is intentionally narrower: one parent Gemma family, paired prefix states, and handoff quality measured by resumed decoding rather than geometric aesthetics.

That narrower framing is a strength. It allows the experiments to distinguish three different possibilities:

1. the regimes are too similar to justify the project;
2. the regimes are separated, but only local fit is learnable;
3. the regimes are separated and a learned intervention can actually recover behavior.

The completed results are most consistent with the third case, but only at a specific intervention site.

4 Problem formulation

Let $r \in \{\text{low}, \text{full}\}$ denote the reduced and fuller regimes. For prompt prefix $x_{1:T}$, layer ℓ, and token position t, define the residual-stream state

$$z_{\ell,t}^{(r)} \in \mathbb{R}^d.$$

The primary objective is to learn a transport map that predicts the fuller-regime state from the reduced-regime state at a chosen intervention site:

$$\hat{z}_{\ell,t}^{(\text{full})} = f_\theta\Big(z_{\ell,t}^{(\text{low})}, c\Big) = z_{\ell,t}^{(\text{low})} + \Delta_\theta\Big(z_{\ell,t}^{(\text{low})}, c\Big),$$

where c is optional conditioning metadata such as layer index, token position, and regime descriptor.

The project began with broad layer-conditioned translators, but the later evidence showed that targeting a specific site matters more than fitting a globally smooth map.

4.1 Training objective

For residual uplift, the objective family is the weighted combination

$$\mathcal{L} = \lambda_{\text{state}}\mathcal{L}_{\text{state}} + \lambda_{\text{logit}}\mathcal{L}_{\text{logit}} + \lambda_{\text{ce}}\mathcal{L}_{\text{ce}},$$

with

$$\mathcal{L}_{\text{state}} = \left\|\hat{z}_{\ell,t}^{(\text{full})} - z_{\ell,t}^{(\text{full})}\right\|_2^2 + \lambda_{\cos}\left(1 - \cos\Big(\hat{z}_{\ell,t}^{(\text{full})}, z_{\ell,t}^{(\text{full})}\Big)\right),$$

and

$$\mathcal{L}_{\text{logit}} = \text{KL}\Big(p_{t+1}^{(\text{full})} \,\Big\|\, p_{t+1}^{(\text{uplift})}\Big).$$

The short-horizon cross-entropy term is

$$\mathcal{L}_{\text{ce}} = -\log p_{t+1}^{(\text{uplift})}\left(y_{t+1}^{(\text{full})}\right),$$

where the target token $y_{t+1}^{(\text{full})}$ is taken from the fuller-regime next-token decision under the same prefix.

In the completed fixed-site ablation at `layer34,last1`, the best current setting is

$$\lambda_{\text{state}} = 1.0, \quad \lambda_{\text{cos}} = 0.2, \quad \lambda_{\text{logit}} = 0.3, \quad \lambda_{\text{ce}} = 1.0,$$

corresponding to the `short-horizon` objective reported later in Table 5.

The deployment criterion is behavioral rather than geometric: improved continuation after handoff. Let $y_{t+i}^{(\text{full})}$ denote greedy continuation under the fuller regime and $\hat{y}_{t+i}$ the continuation after patching the reduced regime and resuming under the fuller regime. For horizon $H \in \{1, 4, 8, 16\}$, define

$$\text{Match@}H = \frac{1}{NH}\sum_{n=1}^{N}\sum_{i=1}^{H}\mathbf{1}\left[\hat{y}_{t+i}^{(n)} = y_{t+i}^{(\text{full},n)}\right].$$

The project only counts as successful when these continuation metrics improve over the proper no-patch baseline.

5 Experimental design

5.1 Paired-state extraction

For each prompt prefix, the same parent Gemma 3n family is run under two operating modes: a reduced E2B-like regime and a fuller E4B regime. The extractor records matched layer/token metadata and paired tensors at aligned positions. Each sample contains:

- prompt identifier;
- token position;
- layer index;
- residual-stream state in both regimes;
- optional KV cache slices;
- optional sparse logits snapshots.

The current paper focuses on residual uplift first. KV uplift remains a secondary object because direct cache prediction is more brittle and was not yet shown to improve the central behavioral metrics.

5.2 Regime construction

The current paper does not treat the two operating regimes as vague labels. They are instantiated by concrete activation and KV-sharing controls exposed through the Gemma runner. Table 1 summarizes the low/full settings used in the completed sanity and pilot runs.

Regime	Activation sparsity	KV shared layers	Scale correction	Active index
low	0.95	20	False	0
full	0.00	0	True	0

Table 1: Low/full regime controls used for the current RegimeLift study. These settings define the reduced E2B-like path and the fuller E4B path used throughout the paired-state pipeline.

Operationally, these controls do the following in our runner. `activation_sparsity` sets the fraction of FFN channels disabled per layer, so higher values produce a narrower active path. `KV shared layers` enables cache-sharing behavior in later blocks of the reduced regime, which changes the attention-state object that must be handed off. `scale correction` toggles a regime-specific normalization correction used by the text path in this implementation. `active index` selects the active stream when Gemma emits multiple stream slots; we keep it fixed at index 0 for all runs in this paper.

5.3 Model families and controls

The completed experiments cover:

- `identity`: no correction;
- `mean-delta`, global linear, and per-layer linear baselines;
- `lowrank` and `mlp` residual translators;
- same-site exact-substitution reference patches (formerly called "oracle" rows in earlier notes).

The strictest control is `identity`. If identity differs from the proper no-patch path, the evaluator is invalid. That check now passes exactly.

5.4 Live handoff implementation

Live evaluation uses a fixed patch procedure. For each prompt prefix, we run a low-regime prefill and extract the residual state at the intervention site. The learned model predicts a replacement state for that site, and a forward hook patches the selected decoder layer output during prefill.

In the main result configuration, the patch is rank-1 in sequence dimension ("last1"): only the final prefix token at the chosen layer is overwritten. For `layer34,last1`, the patch tensor is applied to layer 34, token position $t = T$, after the layer output is produced and before final-logit projection.

After patched prefill, decoding resumes under the full regime using the resulting cache; continuation metrics compare this path against native full-regime continuation on the same prompt. Identity control uses the same code path but patches with the low state itself, which is why identity must match no-patch exactly.

5.5 Prompt families and split protocol

The current evidence is built on synthetic but structured prompt families rather than unconstrained open-ended traces. The prompt pools include arithmetic, symbolic, logical, sequential, and constrained-completion templates. This was a deliberate design choice: the first job of the project was to establish whether the handoff premise is even falsifiable under cleanly paired conditions.

The completed tables in this draft therefore combine:

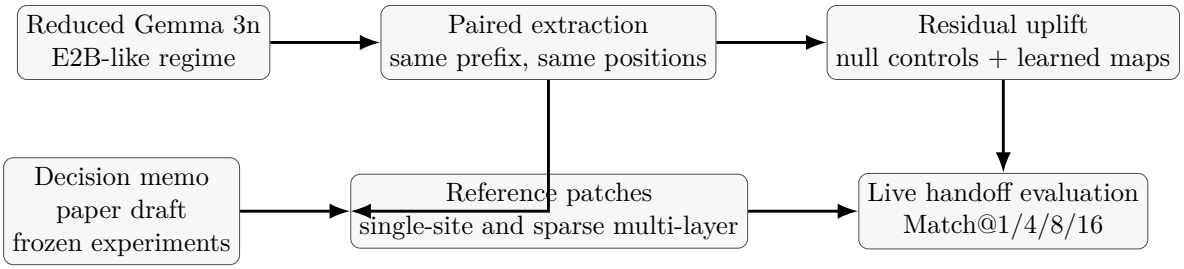

Figure 1: Active RegimeLift pipeline. The work progresses from paired extraction to residual uplift and then to live continuation handoff.

- a pilot split used for fast kill-tests and site sweeps;
- a smaller held-out split used in the frozen targeted-site study;
- a completed heldout80 split used for larger-sample targeted-site validation.

The important discipline is that architectural changes are not justified until they survive the held-out handoff metric, not just the pilot tables.

5.6 Evaluation ladder

The evaluation hierarchy is intentionally brutal:

1. low/full separation sanity checks;
2. extractor alignment checks;
3. null controls and broad learned baselines;
4. live layer sweeps;
5. suffix-span sweeps;
6. targeted-site training and held-out validation.

This sequence matters because it eliminates false-success paths. The project explicitly rejects the claim "the translator works" if only local MSE improves while continuation remains unchanged.

6 Experiment sequence

The evidence in this paper is the output of a fixed sequence of kill-tests rather than one monolithic run. Table 2 records that sequence.

Phase	What was tested	Outcome
0	Low/full separation sanity check	Passed: the two regimes are meaningfully different.
1	Extractor validation on structured prompts	Passed: paired residuals, caches, and logits line up.
2	Null controls plus broad baseline table	Passed: identity equals no-patch; broad MLP is not enough.
3	Layer sweep over candidate sites	Passed: layer 34 is the strongest learned site.
4	Suffix-span sweep at the selected sites	Passed: last1 is already the informative span.
5	Targeted study with held-out prompts	Passed: layer34,last1 generalizes; oracle is reference.
6	Held-out80 rerun with fail-fast checks	Completed: 80 prompts loaded and evaluated.

Table 2: Chronology of the completed experiment program and the gate answered by each phase.

The key transition in that sequence was from broad training to targeted intervention. Broad residual training improved local state metrics but did not recover behavior strongly enough. The subsequent layer and span sweeps isolated the intervention surface before targeted training was attempted.

7 Results

7.1 Regime separation sanity

Before any uplift training, the project must establish that the reduced and fuller regimes are meaningfully different. Table 3 and Figure 2 summarize that first gate using the completed sanity report.

Metric	Value
Prompt count	20
Residual cosine mean	0.5042
Residual MSE mean	51.3558
Next-token KL(full || low)	11.9936
Next-token top-1 agreement	0.0344

Table 3: Global low/full separation metrics from the completed sanity report. These values establish that the handoff problem is real before any translator is trained.

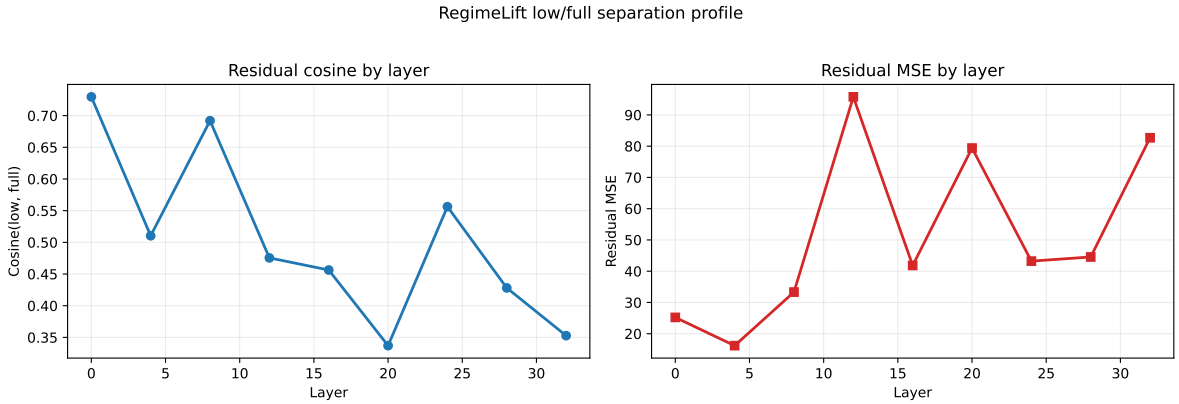

Figure 2: Per-layer low/full separation profile. The mismatch is not uniform across the stack, which is one reason later intervention-site sweeps are necessary.

The important facts here are straightforward. The mean residual cosine is near 0.50 rather than near 1.0, the mean next-token KL is large, and top-1 agreement is low. This is enough to rule out the trivial failure mode in which low and full regimes are operationally identical under the current runner.

7.2 Core targeted-site result

The strongest completed result is the targeted site study summarized in Table 4 and Figure 3. The contrast is decisive:

- `identity` matches `no patch` exactly;
- broad MLP training improves over the no-patch baseline only weakly;
- targeted MLP at `layer16,last1` fails to generalize;
- targeted MLP at `layer34,last1` produces the first strong held-out continuation recovery result.

Table 4: Heldout80 continuation match versus full-regime decoding for the completed targeted-site study. The split contains 80 prompts: arithmetic (21), logic (19), symbolic (17), sequence (12), and constrained generation (11).

Method	h1	h4	h8	h16	Δh8	Δh16
No patch	0.037500	0.025000	0.017188	0.013281	0.000000	0.000000
Identity	0.037500	0.025000	0.017188	0.013281	0.000000	0.000000
Broad MLP	0.075000	0.028125	0.018750	0.010937	0.001562	-0.002344
Targeted MLP @ layer16	0.012500	0.006250	0.010937	0.007812	-0.006250	-0.005469
Targeted MLP @ layer34	0.775000	0.203125	0.115625	0.064062	0.098438	0.050781
Reference layer16	0.012500	0.006250	0.014063	0.013281	-0.003125	0.000000
Reference layer34	0.387500	0.103125	0.060937	0.036719	0.043750	0.023438
Reference stride	0.237500	0.068750	0.046875	0.028906	0.029687	0.015625

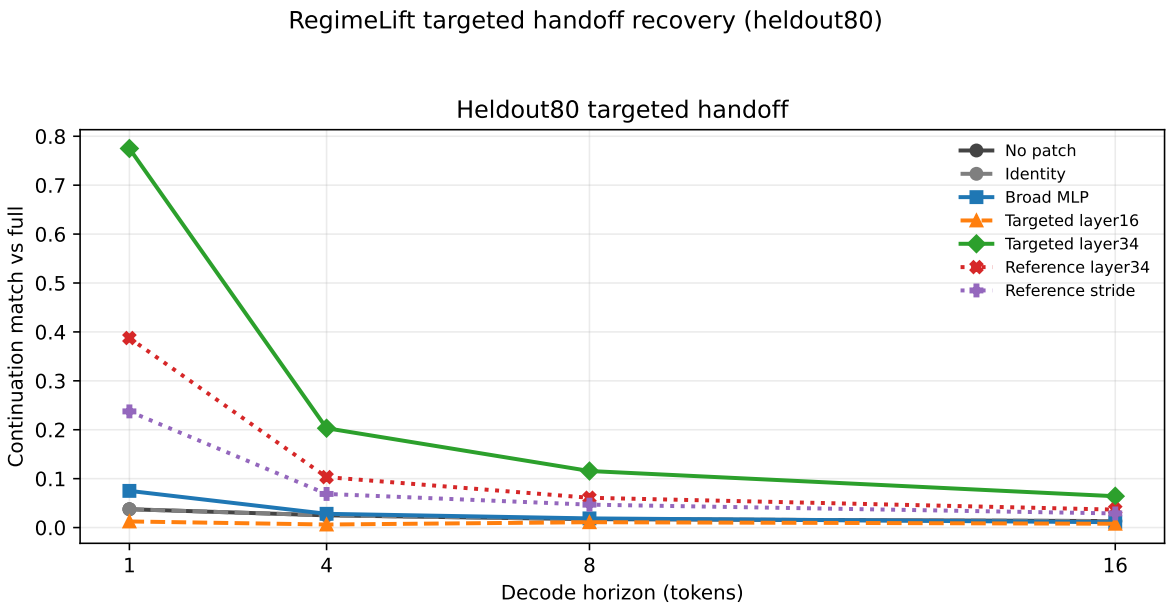

Figure 3: Continuation match curves for the targeted-site study. The practical result is the separation between broad training and targeted training at `layer34,last1`.

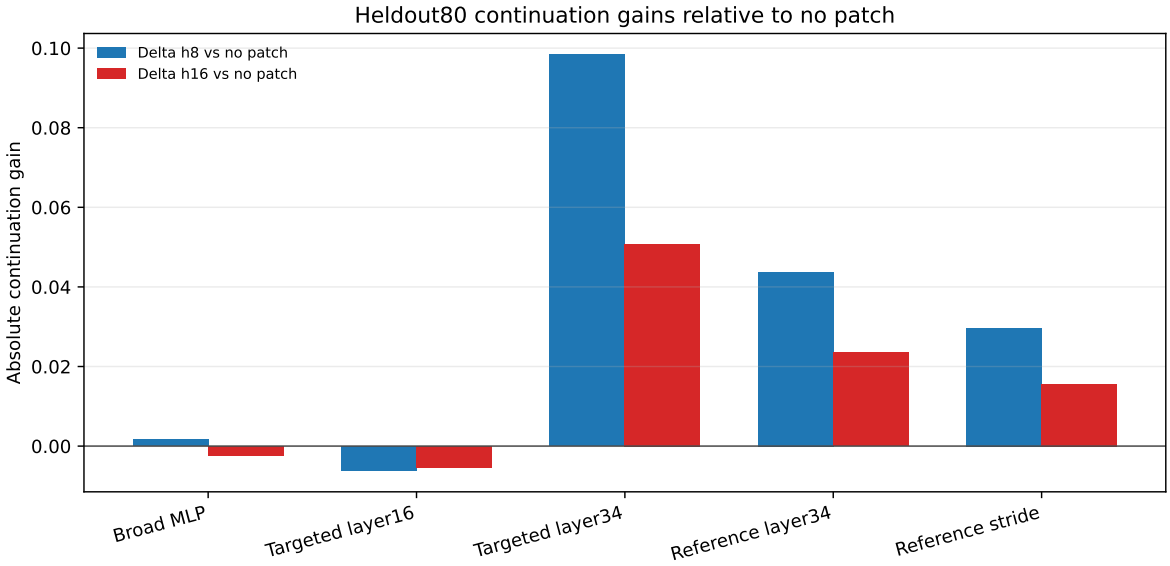

Figure 4: Heldout80 absolute gains over no-patch at horizons 8 and 16. The dominant learned gain comes from the targeted `layer34,last1` model; broad training and the failed `layer16,last1` target do not recover comparable behavior.

The most important row in Table 4 is `Targeted MLP @ layer34`. On the completed heldout80 split it achieves Match@1 of 0.775, Match@4 of 0.203, Match@8 of 0.116, and Match@16 of 0.064. No other learned row comes close. Importantly, `layer16,last1` looked plausible in pilot sweeps and then failed under stronger validation, which is exactly why targeted-site validation was necessary. This is the first point in the project where a learned intervention does what the project is supposed

to do: materially recover continuation behavior on held-out prompts.

7.3 Fixed-site objective ablation

The objective-ablation phase holds the intervention fixed at `layer34,last1` and changes only loss composition. Table 5 exposes the full matrix that was previously summarized in prose: `state-only`, `state+logit`, `heavy-logit`, and `short-horizon`. The ranking is consistent across horizons and deltas: the short-horizon objective is strongest, heavy-logit is second, and state-only is weakest.

Table 5: Heldout80 objective-ablation matrix at fixed `layer34,last1`. Deltas are measured against the same no-patch baseline.

Objective	h1	h4	h8	h16	Δh8	Δh16
State-only	0.312500	0.081250	0.048438	0.028906	0.017188	0.003125
State + logit	0.800000	0.209375	0.118750	0.065625	0.087500	0.039844
Heavy-logit	0.862500	0.225000	0.126562	0.070312	0.095312	0.044531
Short-horizon	0.912500	0.237500	0.132812	0.074219	0.101562	0.048438

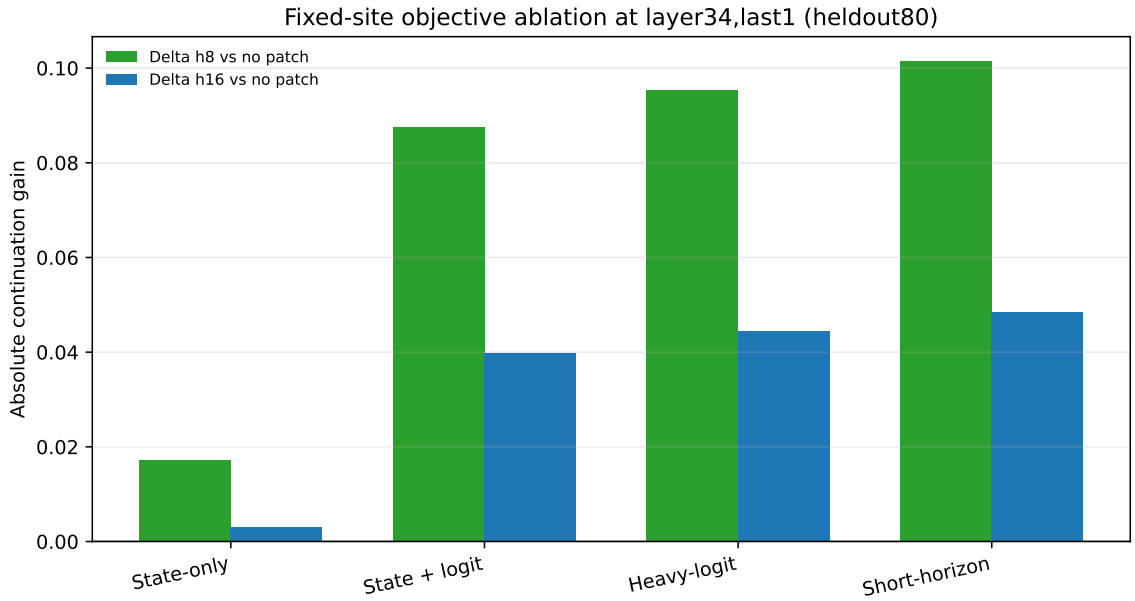

Figure 5: Objective-ablation gains over no-patch at horizons 8 and 16. The short-horizon objective provides the strongest behavioral recovery under a fixed intervention site.

7.4 Full-like versus low-like alignment

Table 6 addresses whether learned uplift is merely perturbing continuation arbitrarily or moving behavior toward fuller-regime decoding. The targeted `layer34` row raises match-to-full-native strongly over no-patch and broad MLP while keeping match-to-low-native low, which is consistent with the intended directional effect. By contrast, broad MLP remains closer to low-native behavior than to full-native behavior.

Table 6: Next-token behavioral alignment at `layer34,last1`: full-native versus low-native.

Method	Match to full-native	Match to low-native	KL(full||method)	KL(low||method)
No patch	0.037500	0.000000	10.364227	15.930509
Reference layer34	0.387500	0.037500	1.197973	9.800478
Broad MLP	0.062500	0.212500	7.455112	3.931631
Targeted layer34	0.287500	0.025000	2.660585	4.552105

7.5 Layer sensitivity

The corrected layer sweep showed that continuation recovery is controlled by a narrow intervention surface. Earlier pilot impressions suggested `layer16` might be the right target. The controlled sweep falsified that as a stable main site and instead showed that the strongest useful region lies substantially later in the stack.

Method	Best layer @ h8	h8	Best layer @ h16	h16
identity	0	0.009375	0	0.012500
mlp	16	0.021875	4	0.020313
oracle_full_state	34	0.037500	34	0.028125

Table 7: Summary of the corrected layer sweep. The learned and reference-patch peaks are not uniform across the stack.

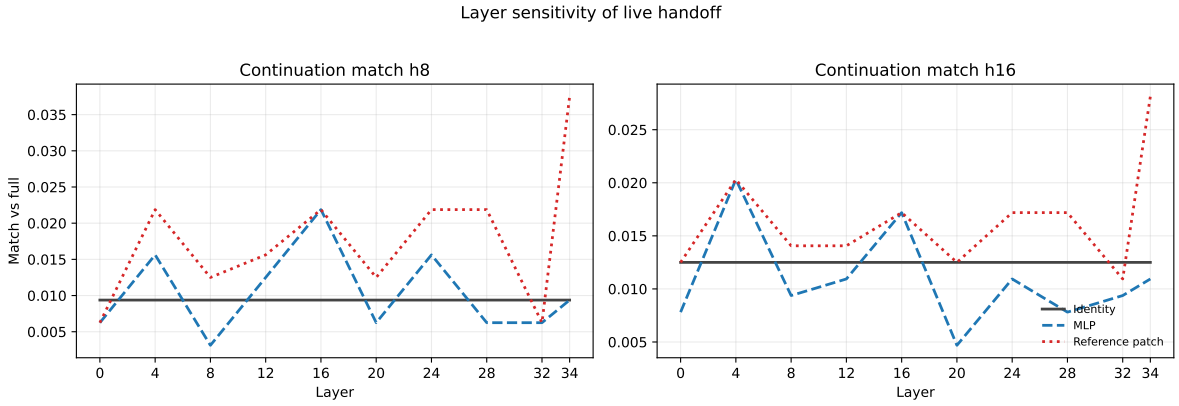

Figure 6: Layer sensitivity of live handoff. The main result is not just that some layers are better than others, but that broad training was washing out the signal at the truly useful site.

7.6 Suffix-span sensitivity

Once the correct intervention site was isolated, the next question was whether the patch needed more recent context than the final prefix token. The completed suffix-span sweep says mostly no: widening from the last token to the last 4 or 8 tokens did not produce a comparable gain to moving the intervention to the right site.

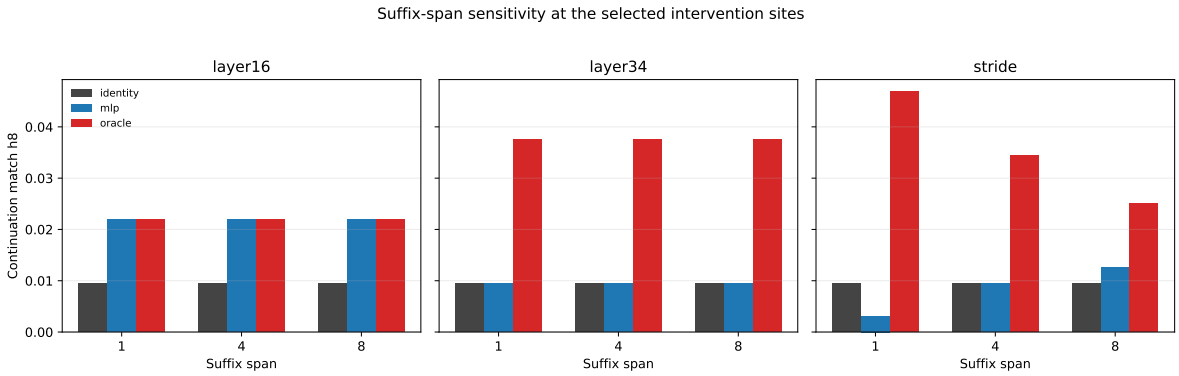

Figure 7: Suffix-span sensitivity at the selected intervention sites. Patch width matters less than intervention site in the completed pilot.

7.7 What the evidence now supports

The current evidence justifies the following claims:

- low/full Gemma 3n operating regimes are measurably distinct;
- the live handoff path is valid because identity equals proper no-patch exactly;
- local residual fit is insufficient as a success criterion;
- the first robust learned handoff site is `layer34,last1`;
- fixed-site objective ablation at `layer34,last1` improves continuation further, with the short-horizon objective reaching $h8 = 0.1328$ and $h16 = 0.0742$ on heldout80;
- same-site exact substitution is a reference patch, not a hard behavioral ceiling;
- intervention placement is a more important bottleneck than broad translator capacity.

8 Discussion

8.1 Why this is not just "two different models"

The right framing is neither extreme. The reduced and fuller regimes are not identical, and the published E2B and E4B artifacts are exposed separately. But for the purposes of this project they are still nested operating regimes of one parent family, trained and structured together rather than unrelated standalone models. That is why same-parent uplift is a coherent hypothesis and why the experiments can legitimately ask whether a small corrective transport suffices. The evidence in this paper suggests that reduced and fuller regimes preserve overlapping task-relevant information while organizing it in different internal coordinates. In that view, successful handoff is best treated as regime-conditioned representational translation rather than simple capacity restoration.

8.2 Why `layer34,last1` changes the project

Before the targeted-site study, the project had evidence of regime separation and local learnability but not of meaningful behavioral recovery. The `layer34,last1` result changes that. It turns the work from "interesting internal geometry" into a practical handoff result with held-out support.

The negative result is equally important. `layer16,last1` looked plausible in earlier sweeps and failed under stronger validation. That failure made the final result more credible because it showed the project was not simply confirming whichever site looked best in an early pilot.

8.3 Why the reference patch is not a ceiling

At some horizons the learned row exceeds the same-site exact-substitution reference. That means raw state fidelity is not identical to optimal rollout behavior. The sensible interpretation is that the reference patch is a control surface, not a mathematical upper bound for continuation match. This paper therefore uses the term *reference patch* consistently instead of *oracle ceiling*.

8.4 Inference relevance

The intended systems use-case is small-first decoding with selective escalation: run the reduced regime when possible, then hand off to fuller activation when continuation quality requires it. This paper establishes the handoff feasibility component, not a full serving policy.

Cost accounting is therefore preliminary. In the completed heldout80 support run at `layer34,last1`, mean low-prefill cost was 283.3 ms, mean full-restart cost was 3191.4 ms, and mean low-plus-handoff pipeline cost was 3504.5 ms, i.e., +313.1 ms slower than full restart in the current instrumentation-heavy setup. That means this paper should not be read as a throughput claim yet; it is a behavioral handoff claim. A production-relevant study still needs trigger policy design, cache-transfer overhead accounting, and clean end-to-end latency measurements under fixed hardware kernels.

8.5 Limitations

An earlier held-out validation accidentally used only 10 prompts. The evaluator was therefore hardened to fail fast on prompt-count mismatch and to write incremental artifacts, and the completed heldout80 rerun is now the canonical larger-sample result.

The other limitation is scope. This paper establishes one strong residual handoff site first. It does not yet claim that the same uplift map works uniformly across intermediate Mix-n-Match regimes, nor that KV uplift is already competitive.

8.6 Threats to validity

Three threats matter most at this stage.

- **Prompt diversity.** The completed heldout80 result is real, but still limited in breadth relative to final publication standards. Additional held-out slices and intermediate-regime pairs remain necessary.
- **Single-site intervention.** RegimeLift has established a first useful site, not a full theory of all intervention surfaces. Sparse multi-layer reference patches remain stronger than many single-site learned interventions.
- **Objective mismatch.** The project already has evidence that excellent local residual fit does not guarantee good continuation recovery. This means future objective changes must be judged behaviorally, not cosmetically.

8.7 Next experimental priority

Fixed-site objective ablation has now been completed at `layer34,last1`. Among `state-only`, `state+logit`, `heavy-logit`, and `short-horizon` objectives, the short-horizon objective is the strongest on heldout80 continuation metrics.

The next high-value experiment is intermediate-regime transfer inside the same parent family (minimum → intermediate, intermediate → full), with the intervention and objective fixed to the best current setting. That experiment determines whether `layer34,last1` is a one-pair result or a broader regime-lift property.

9 Reproducibility and run record

This repository keeps the paper tied to concrete experiment artifacts. Reproducibility rests on three pieces:

- fixed prompt pools with explicit split files;
- explicit run manifests with resolved config paths and prompt counts;
- incremental row-level result dumps for interrupted-run recovery.

The full heldout80 manifest excerpt is provided in Appendix A.

10 Conclusion

RegimeLift now has a stable empirical spine. The project no longer rests on the vague claim that "state translation might work." It now supports a sharper statement:

> Within Gemma 3n's nested E2B/E4B family, low-to-full regime uplift is behaviorally recoverable at a specific learned handoff site, and broad residual fitting alone is not enough to find that site.

That is the main result of the current paper. The next step is not to widen the architecture search blindly. It is to hold the best fixed-site objective at `layer34,last1` and test intermediate-regime generalization within the same Gemma 3n family before expanding to KV-level or flow-based methods.

References

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[3] Aditya Kusupati et al. Matryoshka representation learning. arXiv:2205.13147, 2022. Nested representation learning as a related idea for submodel reuse.

[4] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017.

A Heldout80 Run Manifest Excerpt

```
resolved_heldout_extract_config = .../configs/extract_killtest40_holdout80.yaml
resolved_heldout_prompts_path = .../data/prompt_pool_v1_holdout80.jsonl
"heldout_prompt_count": 80
"phase": "heldout:complete"
"num_rows": 14
"heldout_prompt_id_first": "sym_0253"
"heldout_prompt_id_last": "con_0079"
[targeted] split=heldout prompts=80
Identity control passed
layer34,last1 is the useful learned site
```

Table 8: Manifest excerpt for the completed heldout80 validation run.